Sheik Muhammad Saadiq

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.99
Sri Medha V Junior College	Class XII (Telangana Board of Intermediate Education)	2024	97.00%
Sri Chaitanya Techno H S Tolichowki	Class X (Board of Secondary Education, Telangana)	2022	9.70%

ACHIEVEMENTS

Competitive Programming

- Achieved a **Codeforces rating of 1526.**

KLA AI Hackathon -- Top 5

- Ranked among the **Top 5** teams in the **KLA** AI Hackathon.
- Built an image restoration pipeline using the **SwinIR Transformer** architecture.
- Achieved **89.75% restoration accuracy**, the highest score among all participating teams.

Academic Excellence Award -- IIT Hyderabad

- Awarded for securing the **highest CGPA** in the Chemical Engineering department during the first semester.

Deutsche Börse Hackathon -- 2nd Place

- Secured **2nd place in a Capture The Flag (CTF)** style cybersecurity competition organized by **Deutsche Börse.**

Motive GenAI Hackathon -- Top 10

- Ranked among the **Top 10** teams in a Generative AI hackathon conducted by Motive.
- Developed an image retrieval system capable of returning relevant images from natural language queries.
- Built a **Retrieval-Augmented Generation (RAG) pipeline using vector embeddings** and similarity search for **semantic image retrieval.**

Entrepreneurship

- **Co-founded swapR**, a mobile app designed to simplify train seat exchanges.

EXPERIENCE

Co-Founder & Lead Developer -- swapR

- Co-founded a platform aimed at simplifying train seat exchanges.
- Served as the **sole developer,** leading the complete technical development of the product.
- Designed and developed the mobile application using **Flutter and FastAPI.**

Core Member -- Lambda Software Development Club, IIT Hyderabad

- Contributed to technical initiatives and software development activities within the club.
- Collaborated with student teams and **college administration** on coding events and development projects.

SKILLS

Programming Languages - Go, Python, C++, Dart, JavaScript, SQL

Frameworks & Libraries - Flutter, FastAPI, PyTorch, NumPy, React

Developer Tools - Docker, Git, Linux, Bash

Areas of Interest - Backend Engineering, Machine Learning, Competitive Programming, Mobile Application Development

PROJECTS

Edmit -- NEET UG College Analytics Platform

Tech Stack: Python, PostgreSQL, JavaScript

- Built a full-stack platform for analyzing NEET UG admission trends and cutoff data.
- Collected, cleaned, and normalized admission records from official government counseling portals.
- Aggregated data for 750+ colleges across multiple states, categories, and quotas.

Chess Engine

Tech Stack: C++

- Implemented a chess engine core from scratch using bitboards for efficient board representation and move generation.
- Developed complete legal move generation covering castling, en passant, promotions, checks, pins, and check evasions.
- Achieved >20M nodes/sec search performance.

AutoGrad

Tech Stack: Python

- Implemented an automatic differentiation engine from scratch inspired by PyTorch's autograd.
- Developed computational graph construction and reverse-mode backpropagation.
- Used the framework to train simple neural networks and understand deep learning internals.

IITH Mess Registration Portal -- Institute-wide Mess Registration Platform

Tech Stack: Flutter, Backend Development

- Contributed to both the mobile application and backend components of the mess registration system.

AnilistStream -- Anime Metadata Aggregation and Streaming Backend

Tech Stack: Go, Python, Docker

- Built a backend service that discovers anime episodes from multiple providers through automated web scraping.

RELEVANT COURSES

Programming Courses - Introduction to Programming in C, Introduction to AI/ML

Mathematics - Linear Algebra, Applied Mathematics, Numerical Methods, Calculus, Differential Equations